

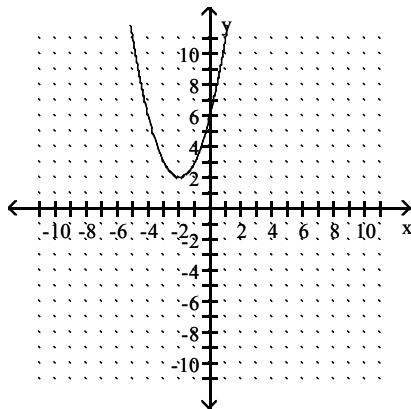
Name _____

SHORT ANSWER. Write the word or phrase that best completes each statement or answers the question.**Use the given conditions to write an equation for the line in the indicated form.**

- 1) Passing through (5, 4) and perpendicular to the line whose equation is $y = 7x + 7$; point-slope form 1) _____
- 2) Passing through (5, -2) and parallel to the line whose equation is $y = -9x + 8$; slope-intercept form 2) _____

MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.**The graph of a quadratic function is given. Determine the function's equation.**

- 3) _____



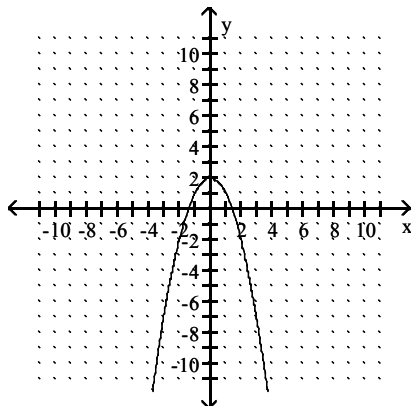
A) $f(x) = (x + 2)^2 + 2$

B) $h(x) = (x - 2)^2 + 2$

C) $g(x) = (x + 2)^2 - 2$

D) $j(x) = (x - 2)^2 - 2$

- 4) _____



A) $f(x) = -x^2 - 4x - 4$

B) $j(x) = -x^2 + 2$

C) $g(x) = -x^2 + 4x + 4$

D) $h(x) = -x^2 - 2$

SHORT ANSWER. Write the word or phrase that best completes each statement or answers the question.

Find the coordinates of the vertex for the parabola defined by the given quadratic function.

5) $f(x) = (x + 1)^2 - 7$ 5) _____

6) $f(x) = 3 - x^2 + 2x$ 6) _____

Find the axis of symmetry of the parabola defined by the given quadratic function.

7) $y + 4 = (x + 2)^2$ 7) _____

Find the range of the quadratic function.

8) $f(x) = 11(x - 2)^2 + 8$ 8) _____

Find the x-intercepts (if any) for the graph of the quadratic function.

9) $f(x) = 4 + 5x + x^2$ 9) _____

10) $f(x) = (x + 3)^2 - 9$ 10) _____

11) $f(x) = x^2 + 18x + 58$ Give your answers in exact form. 11) _____

Find the y-intercept for the graph of the quadratic function.

12) $f(x) = -x^2 - 2x + 8$ 12) _____

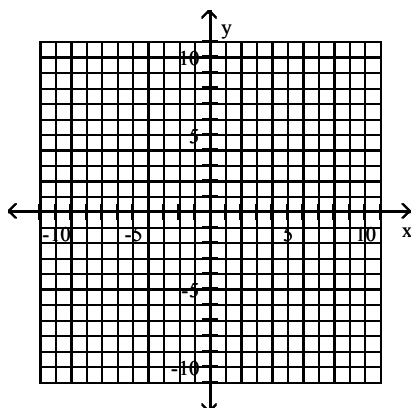
13) $y + 9 = (x - 3)^2$ 13) _____

Find the domain and range of the quadratic function whose graph is described.

14) The vertex is $(-1, -2)$ and the graph opens down. 14) _____

Use the vertex and intercepts to sketch the graph of the quadratic function.

15) $f(x) = -3(x + 5)^2 + 1$ 15) _____



Determine whether the given quadratic function has a minimum value or maximum value. Then find the coordinates of the minimum or maximum point.

16) $f(x) = -x^2 + 2x - 8$ 16) _____

Solve the problem.

17) A developer wants to enclose a rectangular grassy lot that borders a city street for parking. If the developer has 200 feet of fencing and does not fence the side along the street, what is the largest area that can be enclosed? 17) _____

18) The profit that the vendor makes per day by selling x pretzels is given by the function $P(x) = -0.002x^2 + 1.4x - 300$. Find the number of pretzels that must be sold to maximize profit. 18) _____

Determine whether the function is a polynomial function.

19) $f(x) = \sqrt[4]{x^5} - x^4 + 4$ 19) _____

20) $f(x) = \frac{2 - x^5}{9}$ 20) _____

Find the degree of the polynomial function.

21) $f(x) = 4x - x^3 + \frac{4}{3}$ 21) _____

Find the x-intercepts of the polynomial function. State whether the graph crosses the x-axis, or touches the x-axis and turns around, at each intercept.

22) $f(x) = x^4 - 49x^2$ 22) _____

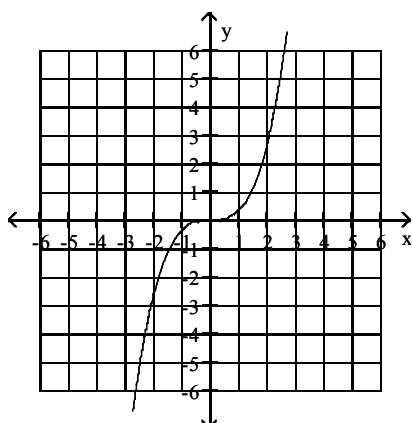
23) $f(x) = x^2(x - 1)(x - 3)$ 23) _____

Find the y-intercept of the polynomial function.

24) $f(x) = -x^2(x + 2)(x^2 - 1)$ 24) _____

Determine whether the graph of the polynomial has y-axis symmetry, origin symmetry, or neither.

25) 25) _____



Use the Leading Coefficient Test to determine the end behavior of the polynomial function.

26) $f(x) = x + 2x^2 - 5x^3$ 26) _____

27) $f(x) = (x + 1)(x + 3)(x + 4)^3$

27) _____

Find the zeros for the polynomial function and give the multiplicity for each zero. State whether the graph crosses the x-axis or touches the x-axis and turns around, at each zero.

28) $f(x) = \frac{1}{2}x^4(x^2 - 3)(x - 1)$

28) _____

29) $f(x) = x^3 + 11x^2 + 39x + 45$

29) _____

Use the Intermediate Value Theorem to determine whether the polynomial function has a real zero between the given integers.

30) $f(x) = 3x^5 + 3x^3 - 4x^2 - 5$; between 1 and 2

30) _____

Determine the maximum possible number of turning points for the graph of the function.

31) $f(x) = x^5 + 2x^6$

31) _____

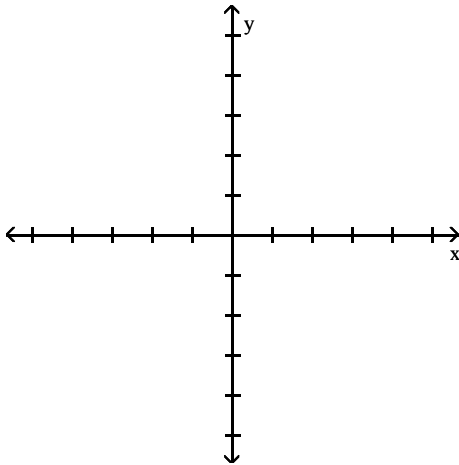
32) $f(x) = x^4(x^4 - 3)(4x - 4)$

32) _____

Graph the polynomial function.

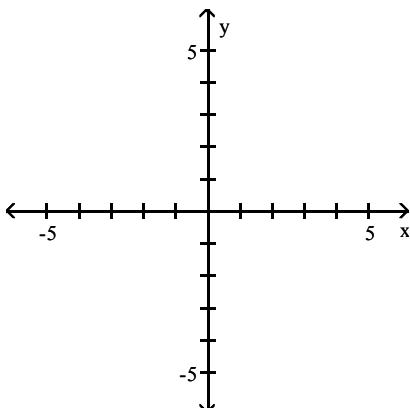
33) $f(x) = x^3 + 8x^2 - x - 8$

33) _____



34) $f(x) = 4x^4 + 4x^3$

34) _____



Answer Key

Testname: MTH_131_PRACTICE_EXAM_4_A

1) $y - 4 = -\frac{1}{7}(x - 5)$

2) $y = -9x + 43$

3) A

4) B

5) $(-1, -7)$

6) $(1, 4)$

7) $x = -2$

8) $[8, \infty)$

9) $(-1, 0)$ and $(-4, 0)$

10) $(0, 0)$ and $(-6, 0)$

11) $(-9 \pm \sqrt{23}, 0)$

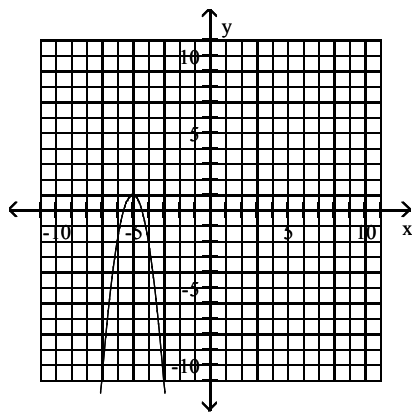
12) $(0, 8)$

13) $(0, 0)$

14) Domain: $(-\infty, \infty)$

Range: $(-\infty, -2]$

15)



16) maximum; $(1, -7)$

17) 5000 ft^2

18) 350 pretzels

19) No

20) Yes

21) 3

22) 0, touches the x-axis and turns around;

7, crosses the x-axis;

-7, crosses the x-axis

23) 0, touches the x-axis and turns around;

1, crosses the x-axis;

3, crosses the x-axis

24) 0

25) origin symmetry

26) rises to the left and falls to the right

27) falls to the left and rises to the right

28) 0, multiplicity 4, touches x-axis and turns around;

1, multiplicity 1, crosses x-axis;

$\sqrt{3}$, multiplicity 1, crosses x-axis;

$-\sqrt{3}$, multiplicity 1, crosses x-axis

Answer Key

Testname: MTH_131_PRACTICE_EXAM_4_A

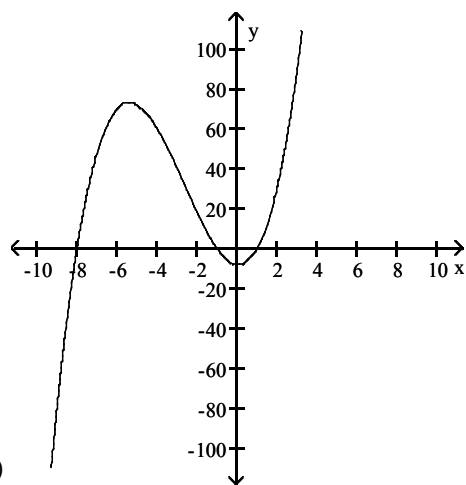
29) -3, multiplicity 2, touches the x-axis and turns around;

-5, multiplicity 1, crosses the x-axis.

30) $f(1) = -3$ and $f(2) = 99$; yes

31) 5

32) 8



33)

34)

